Raft is a consensus algorithm designed as an alternative to Paxos.
It was created by Diego Ongaro and John Ousterhout at Stanford University in 2014.
Raft achieves consensus via an elected leader.
A server in Raft can be in one of three states: Leader, Follower, or Candidate.
In normal operation, there is exactly one leader and all of the other servers are followers.
Followers are passive: they issue no requests on their own but respond to incoming RPCs.
The leader handles all client requests and coordinates log replication across the cluster.
Raft divides time into terms of arbitrary length, numbered with consecutive integers.
Each term begins with an election, in which one or more candidates attempt to become leader.
If a candidate wins the election, it serves as leader for the rest of the term.
Heartbeat messages are periodically sent by the leader to maintain its authority.
If a follower receives no communication over an election timeout, it assumes there is no leader.
The election timeout is randomized between 150ms and 300ms to prevent split votes.
Log entries consist of a state machine command, term number, and an integer index.
Once a log entry created by the leader has been replicated on a majority of servers, it is committed.